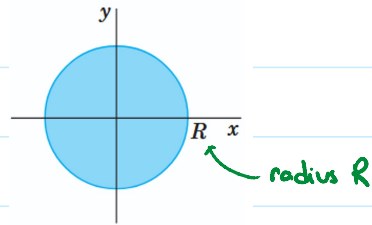


Pre-filled: Circular Membrane

Saturday, July 26, 2025 6:21 PM

13-Laplace's Equation on a circular membrane.

13.1 Suppose our membrane is circular:



Then the 2D wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

can be expressed in polar coordinates:

13.2 We will massively simplify our problem by assuming:

13.3 The problem.

13.4 Separation of variables:

$$\text{Let } u(r, t) = W(r)G(t)$$

Thus: $\ddot{G} + k^2 c^2 G = 0 \Rightarrow G(t) = A \cos(\lambda t) + B \sin(\lambda t)$

By now, we know that $u_t(r, 0) = 0 \Rightarrow B = 0$

so: $G(t) = A \cos(\lambda t)$ where $\lambda = kc$

On the other hand: $\frac{1}{W} W'' + \frac{1}{rW} W' = -k^2 \Rightarrow W'' + \frac{1}{r} W' + k^2 W = 0$

13.5 The only physically viable solution to

$$\frac{d^2 W}{dr^2} + \frac{1}{r} \frac{dW}{dr} + k^2 W = 0$$

is

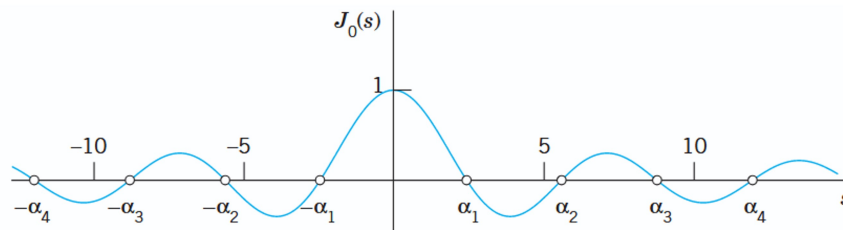


Fig. 308. Bessel function $J_0(s)$.

13.6 So far, we have:

$$u(r, t) = J_0(kr) \cdot A \cos(\lambda t)$$

Now, to satisfy the condition

$$u(R, t) = 0, \text{ we need } J_0(kR) = 0$$

so kR must be a root of the Bessel function $J_0(\cdot)$

in other words:

so we say:

so we say:

Furthermore: $\lambda = cK \Rightarrow$

So we say: $u_m(r, t) =$

13.6 Finally, to satisfy $u(r, 0) = f(r)$
we insist

$$u(r, t) = \sum_{m=1}^{\infty} u_m(r, t) =$$

Then $u(r, 0) = f(r)$ becomes:

These A_m are called the coefficients of the Fourier-Bessel series

$$A_m = \frac{2}{R^2 J_1^2(\alpha_m)} \int_0^R r f(r) J_0\left(\frac{\alpha_m}{R} r\right) dr$$

Values of α_m , $J_1(\alpha_m)$ are known and will be provided to you in a table.

The integral $\int_0^R r f(r) J_0\left(\frac{\alpha_m}{R} r\right) dr$ is usually numerically computed, but for special cases of $f(r)$ we can figure it out.

13.7 Example: Suppose

$$f(r) = 1.108 J_0\left(\frac{2.4048}{2}r\right) - 0.140 J_0\left(\frac{8.6537}{2}r\right)$$

and $u(r, t)$ satisfies $u_{tt} = c^2 \nabla^2 u$, $c = 1$

with BC: $u(2, t) = 0$

IC: $u(r, 0) = f(r)$, $u_t(r, 0) = 0$

on a circular membrane of radius $R = 2$.

What is $u(r, t)$?

m	α_m	$J_1(\alpha_m)$
1	2.40483	0.51915
2	5.52008	-0.34026
3	8.65373	0.27145
4	11.79153	-0.23246
5	14.93092	0.20655
6	18.07106	-0.18773
7	21.21164	0.17327
8	24.35247	-0.16170
9	27.49348	0.15218
10	30.63461	-0.14417

Soln We recognize: $f(r) =$

so we get:

Then

$$\begin{aligned} u(r, t) &= 1.108 \cos\left(\frac{c \alpha_1}{R} t\right) J_0\left(\frac{\alpha_1}{R} r\right) - 0.140 \cos\left(\frac{c \alpha_3}{R} t\right) J_0\left(\frac{\alpha_3}{R} r\right) \\ &= 1.108 \cos\left(\frac{2.40483}{2} t\right) J_0\left(\frac{2.40483}{2} r\right) - 0.140 \cos\left(\frac{8.65373}{2} t\right) J_0\left(\frac{8.65373}{2} r\right) \end{aligned}$$

13.8 Harmonics Frequency

In the solution for $u(r, t)$, λ_m are harmonic wavenumbers, and

$f_m = \frac{\lambda_m}{2\pi}$ are called harmonic frequencies.

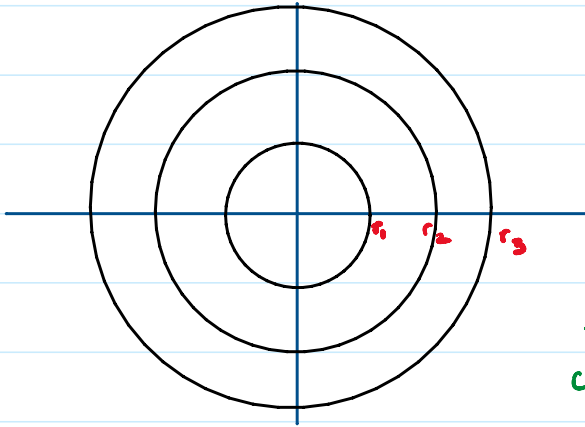
For instance, in the example above, the relevant harmonic frequencies are

$$f_1 = \frac{\lambda_1}{2\pi} = \frac{(c \alpha_1 / R)}{2\pi} = \frac{(2.40483/2)}{2\pi} \approx 0.1914 \text{ Hz}$$

$$f_3 = \frac{\lambda_3}{2\pi} = \frac{(c \alpha_3 / R)}{2\pi} = \frac{(8.65373/2)}{2\pi} \approx 0.6886 \text{ Hz}$$

13.9 Nodal Circles:

$m=3$



inner
circles

For $J_0\left(\frac{\alpha_m}{R}r\right)$, whenever $J_0(\cdot) = 0$ we get a nodal circle.

For example, when $m=3$, there will be three nodal circles:

First one when:

Second one when

Third one when

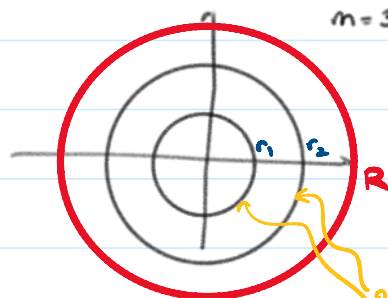
Note: the final circle will always be at radius R , we'll ignore it when discussing nodal circles for harmonics, we only care about the inner circles.

13.10 For the example above, draw the nodal circles for all harmonics and identify the radius of each circle

Soln: We have $u(r, t) = A_1 \cos\left(\frac{\alpha_1}{R}t\right) J_0\left(\frac{\alpha_1}{R}r\right) + A_3 \cos\left(\frac{\alpha_3}{R}t\right) J_0\left(\frac{\alpha_3}{R}r\right)$

when $m=1$, there is no nodal circle for the harmonic (no inner circle)

when $m=3$, there are two nodal circles for the harmonic, with radius:



$m=3$

$$\text{with } r_1 = \left(\frac{\alpha_1}{\alpha_3}\right)R = \left(\frac{2.40483}{8.65373}\right)^2 \approx 0.5558$$

$$r_2 = \left(\frac{\alpha_2}{\alpha_3}\right)R = \left(\frac{5.52008}{8.65373}\right)^2 \approx 1.2758$$

2 nodal
circles



2 nodal
circles

